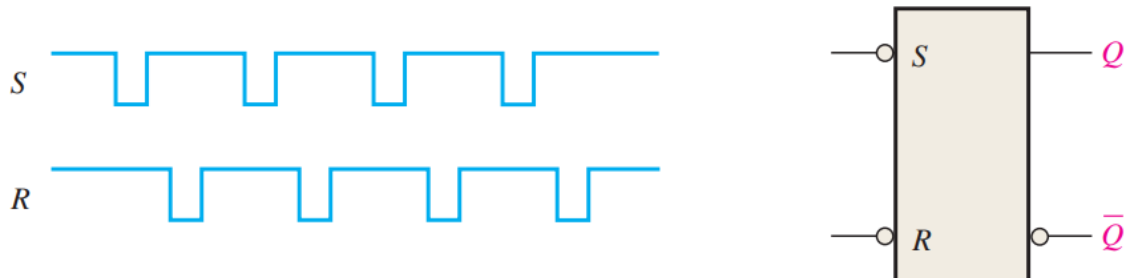


Chapter 7

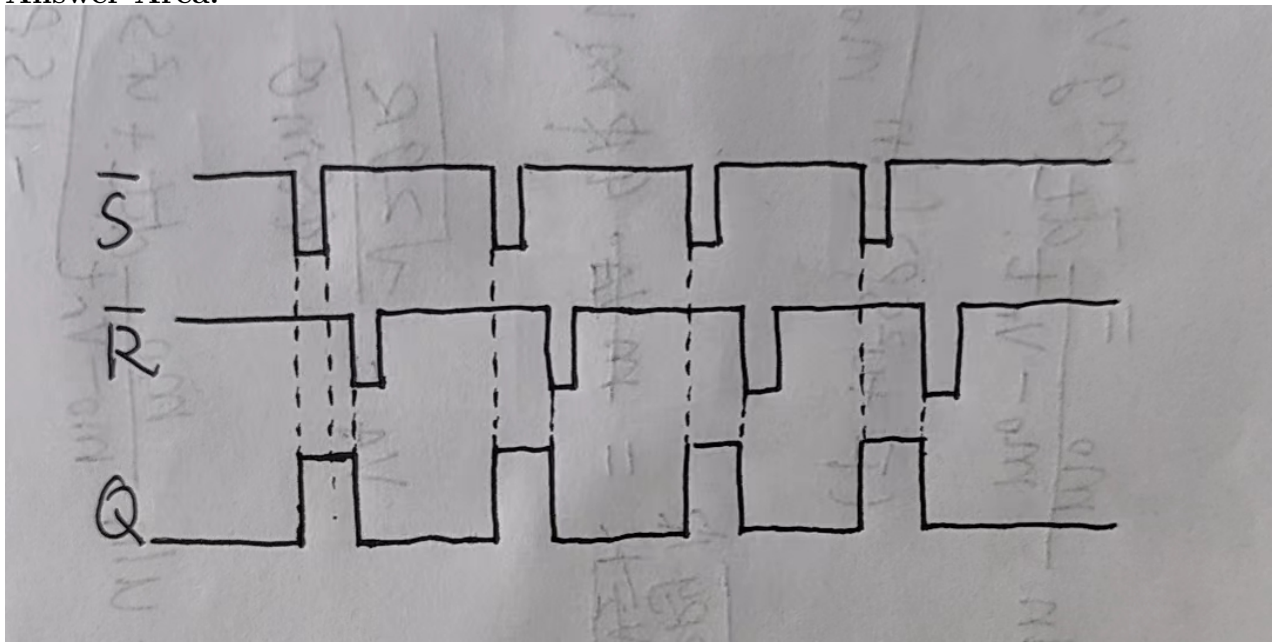
2024.2.22

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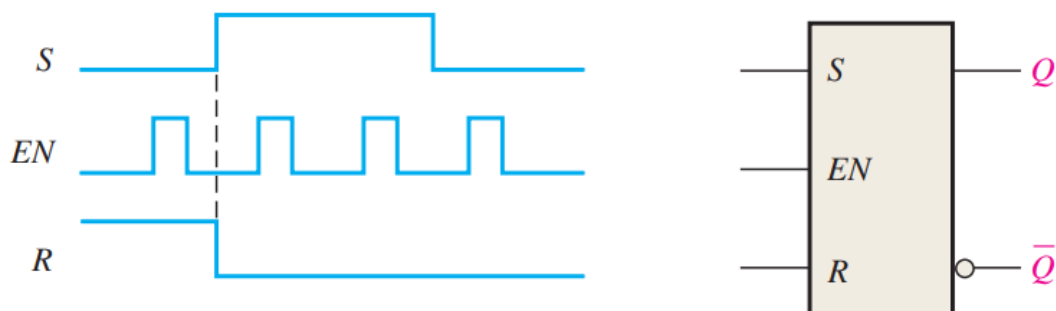
1. If the waveforms in Figure 7-70 are applied to an active-HIGH S-R latch, draw the resulting Q output waveform in relation to the inputs. Assume that Q starts LOW.



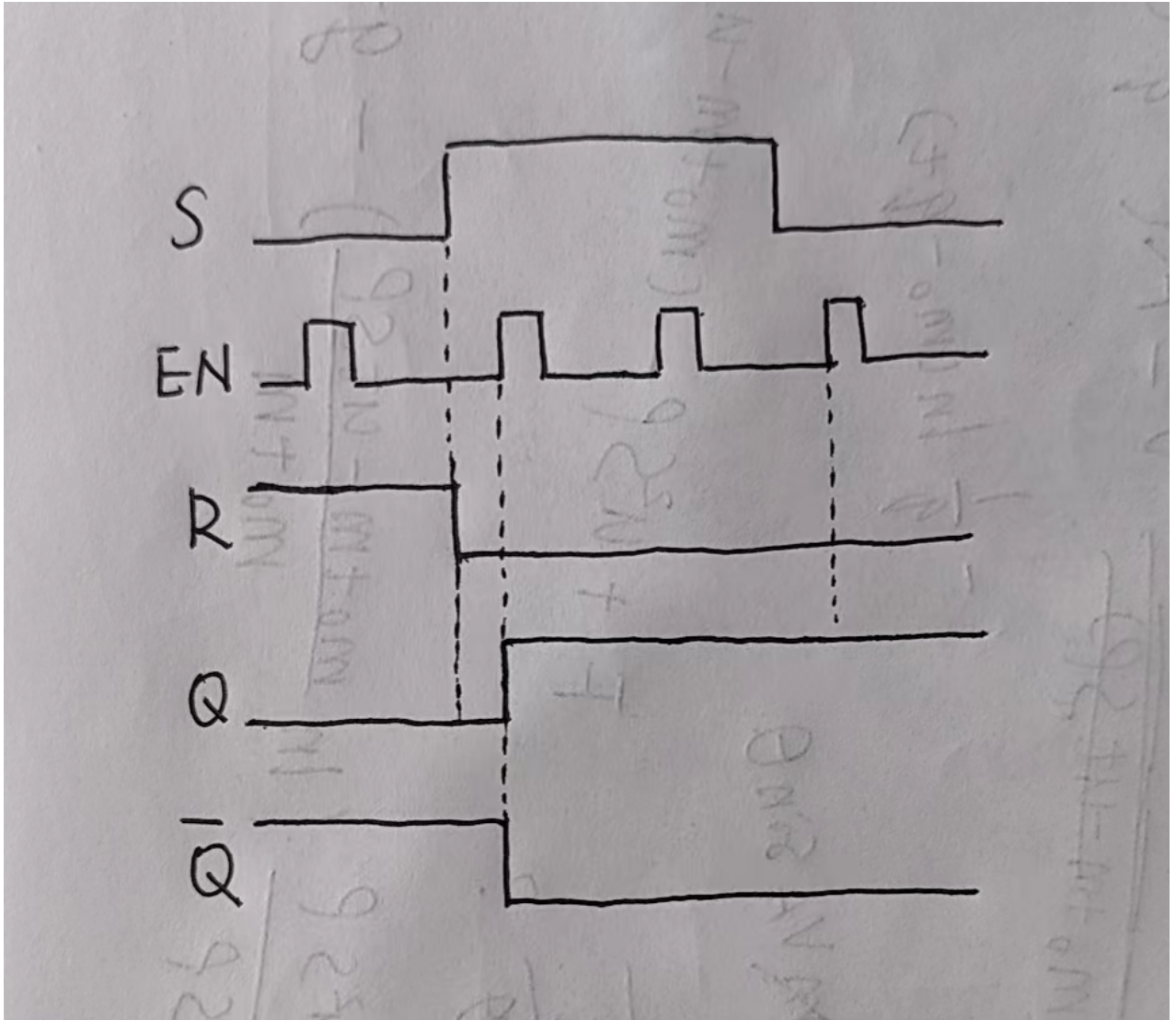
Answer Area:



4. For a gated S-R latch, determine the Q and Q outputs for the inputs in Figure 7-73. Show them in proper relation to the enable input. Assume that Q starts LOW.



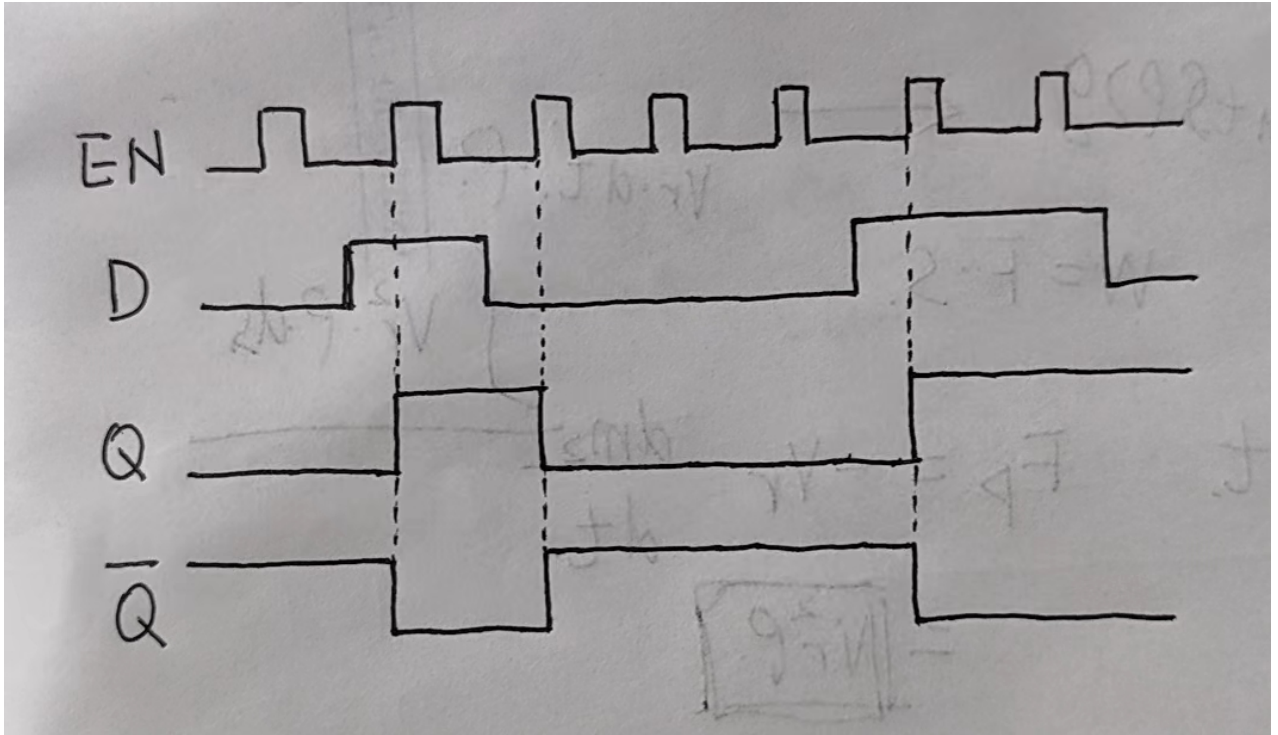
Answer Area:



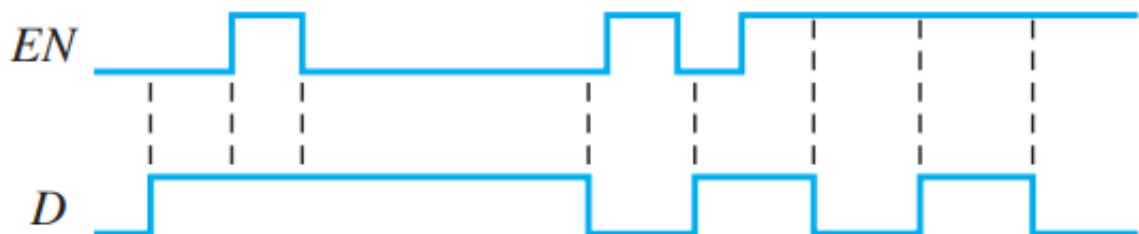
5. Determine the output of a gated D latch for the inputs in Figure 7-74.



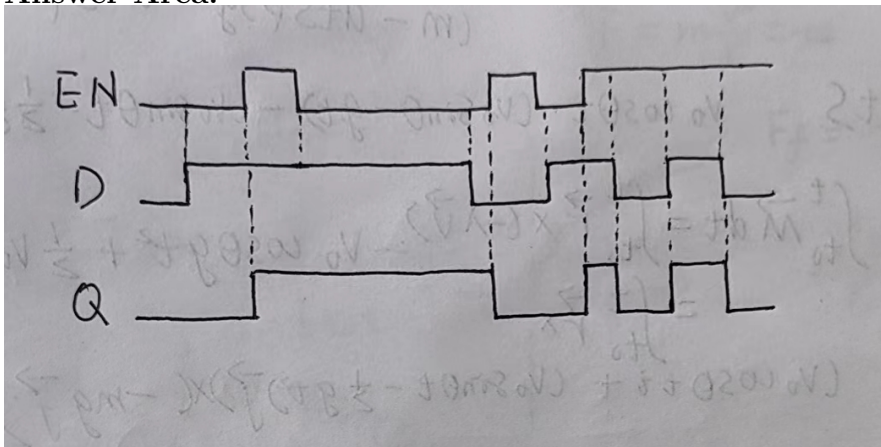
Answer Area:



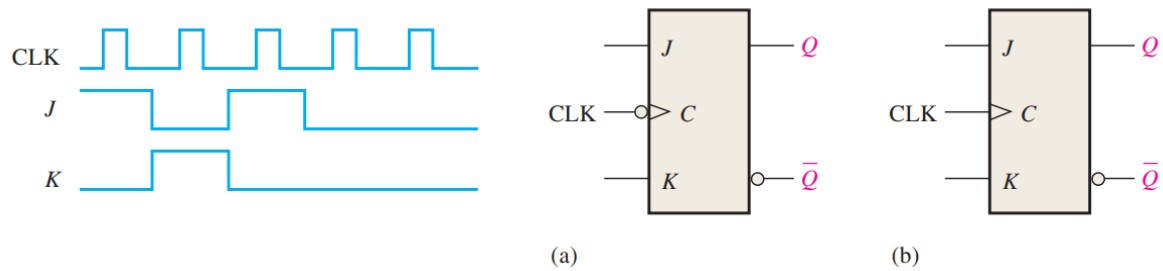
7. For a gated D latch, the waveforms shown in Figure 7-76 are observed on its inputs. Draw the timing diagram showing the output waveform you would expect to see at Q if the latch is initially RESET.



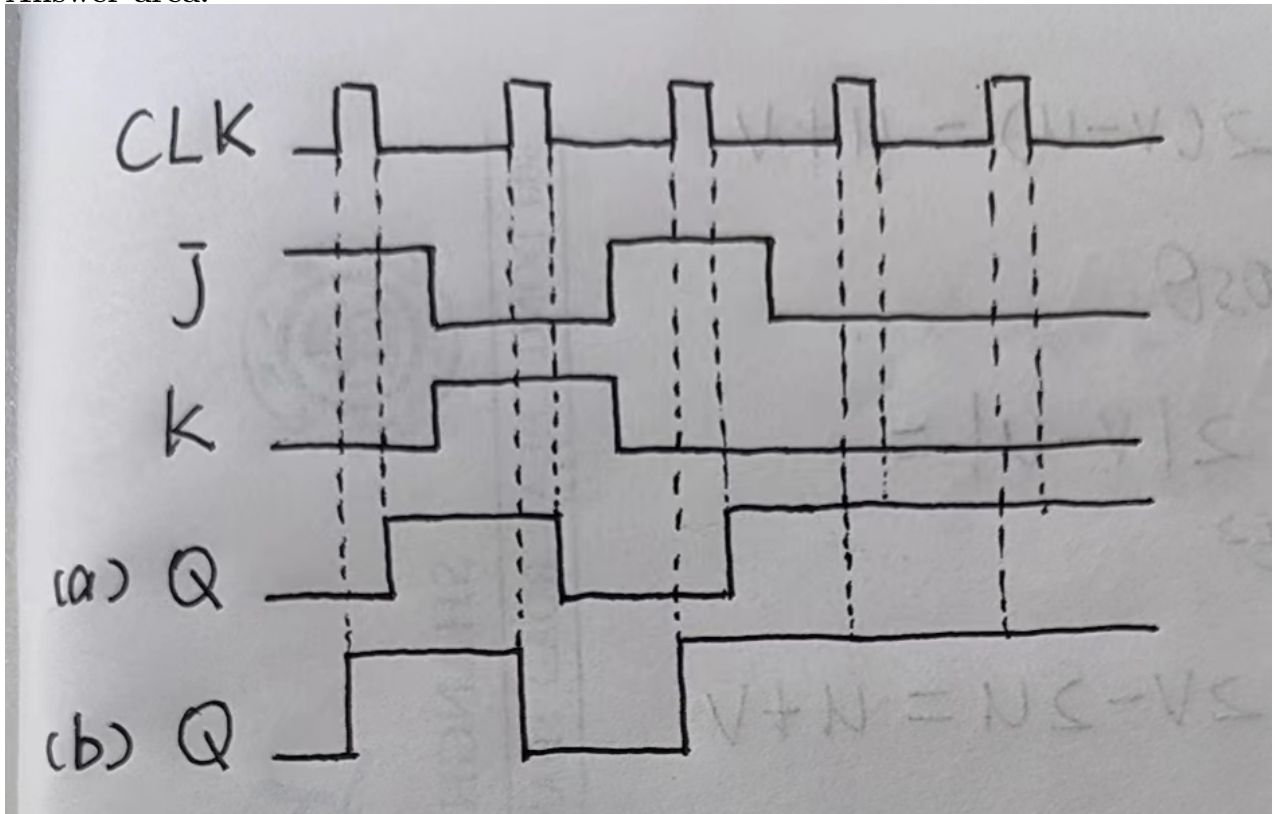
Answer Area:



8. Two edge-triggered J-K flip-flops are shown in Figure 7-77. If the inputs are as shown, draw the Q output of each flip-flop relative to the clock, and explain the difference between the two. The flip-flops are initially RESET.

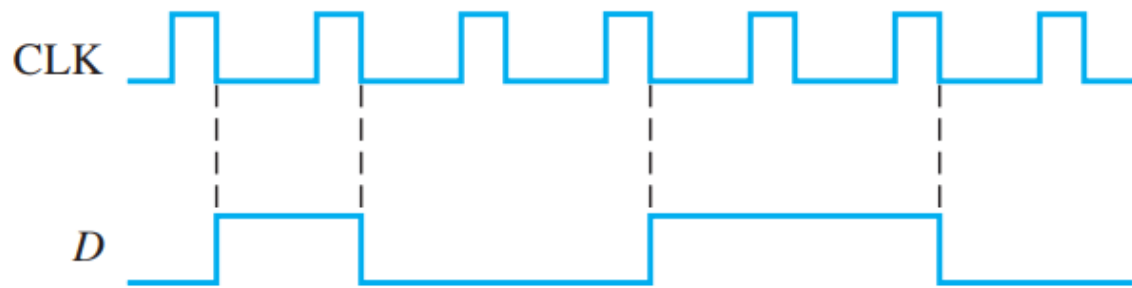


Answer area:

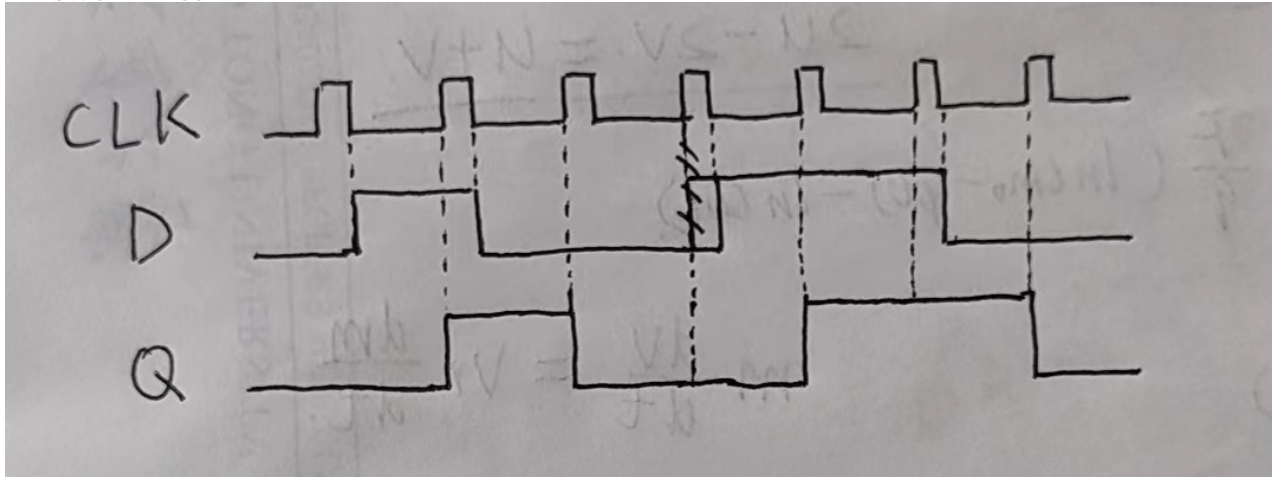


For the flip-flop in (a), the output change only at the falling edge of the clock pulse; for the flip-flop in (b), the output change only at the rising edge of the clock pulse.

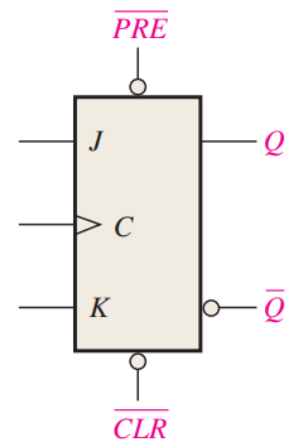
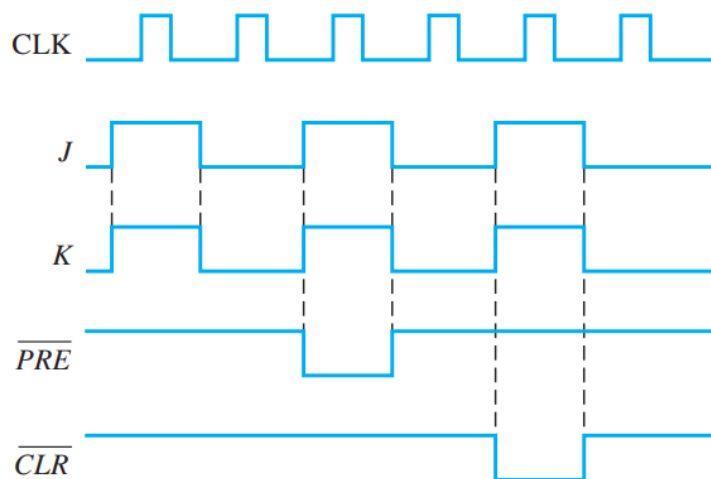
10. Draw the Q output relative to the clock for a D flip-flop with the inputs as shown in Figure 7-79. Assume positive edge-triggering and Q initially LOW.



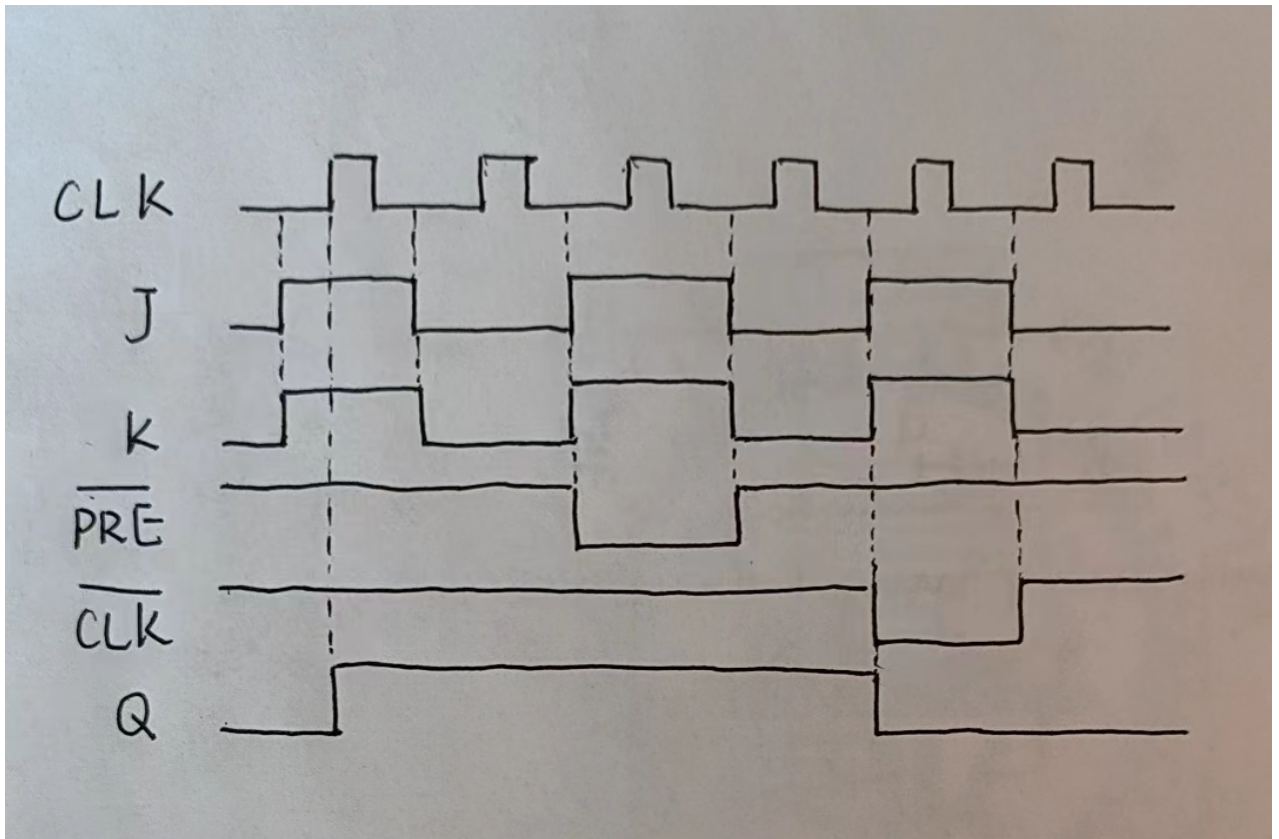
Answer Area:



14. Determine the Q waveform relative to the clock if the signals shown in Figure 7-83 are applied to the inputs of the J-K flip-flop. Assume that Q is initially LOW.

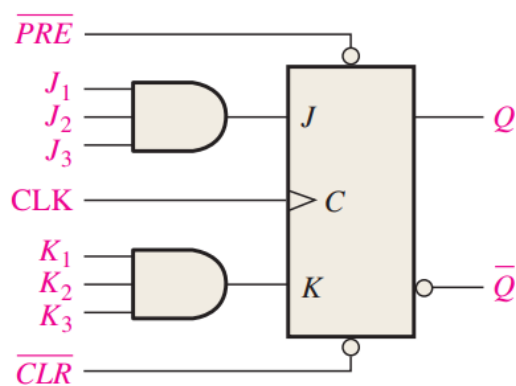


Answer Area:



16. The following serial data are applied to the flip-flop through the AND gates as indicated in Figure 7-85. Determine the resulting serial data that appear on the Q output. There is one clock pulse for each bit time. Assume that Q is initially 0 and that PRE and CLR are HIGH. Rightmost bits are applied first.

$J_1 : 1010011 \quad J_2 : 0111010 \quad J_3 : 1111000$
 $K_1 : 0001110 \quad K_2 : 1101100 \quad K_3 : 1010101$



Answer Area:

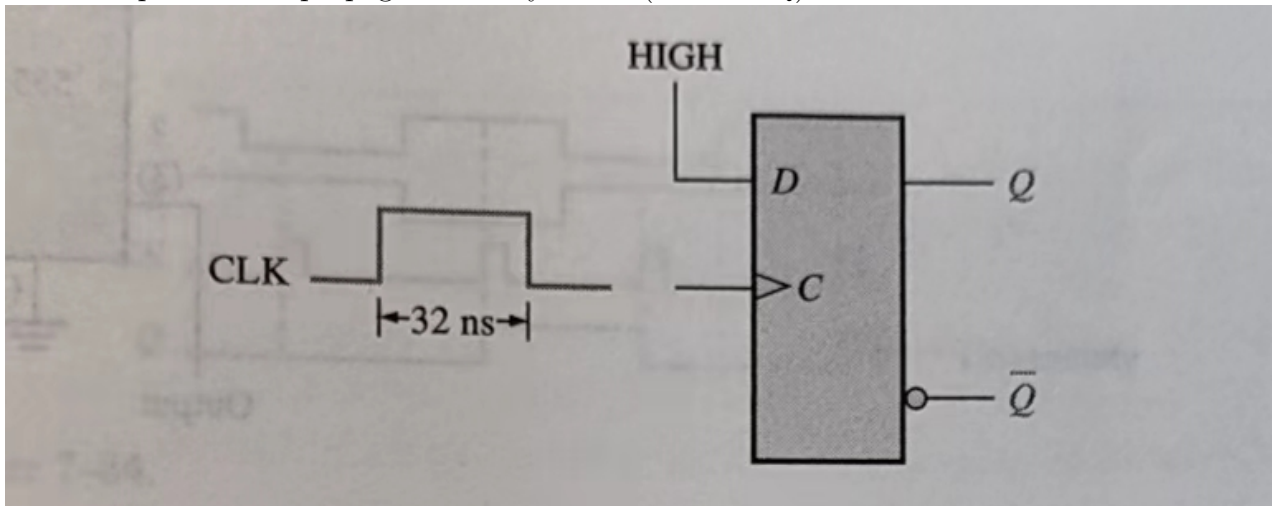
$J = J_1 J_2 J_3 : 0010000$

$K = K_1 K_2 K_3 : 0000100$

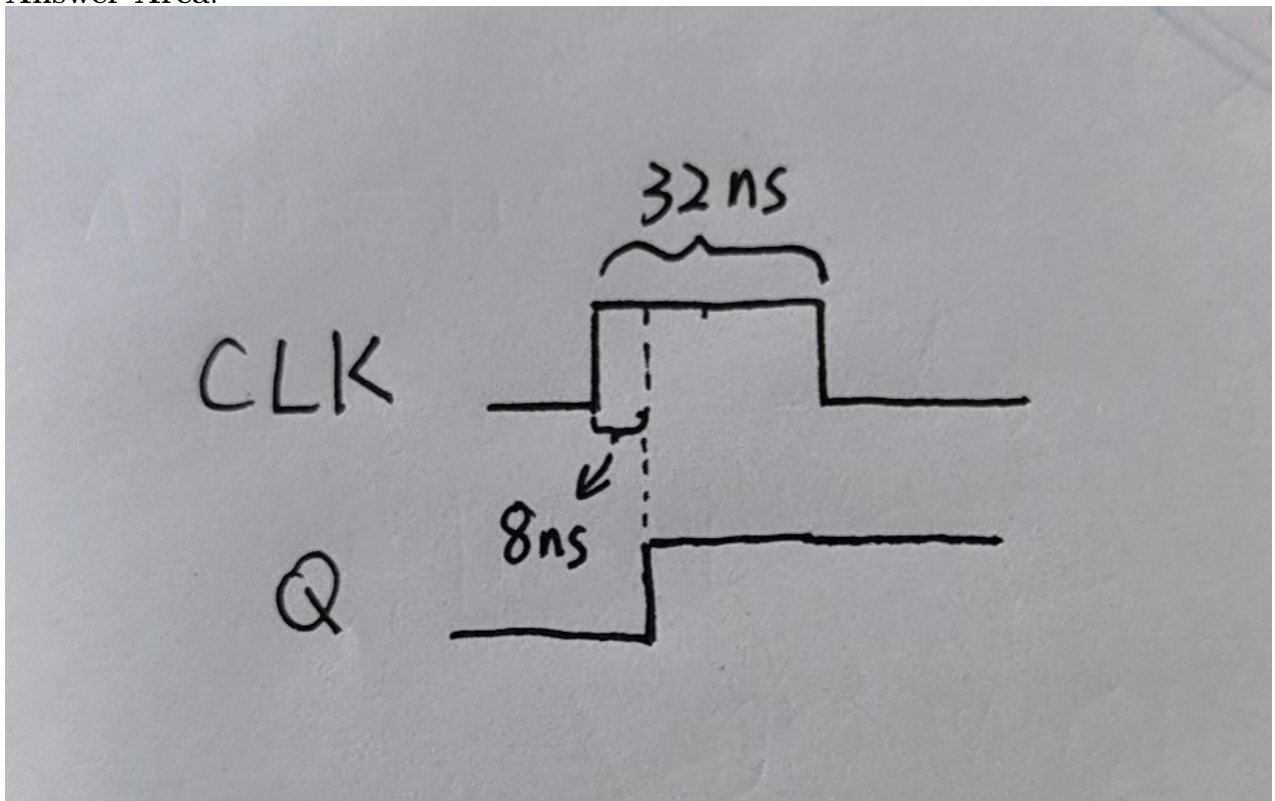
Because there is one clock pulse for each bit time.

Therefore the Q output is: 0011000.

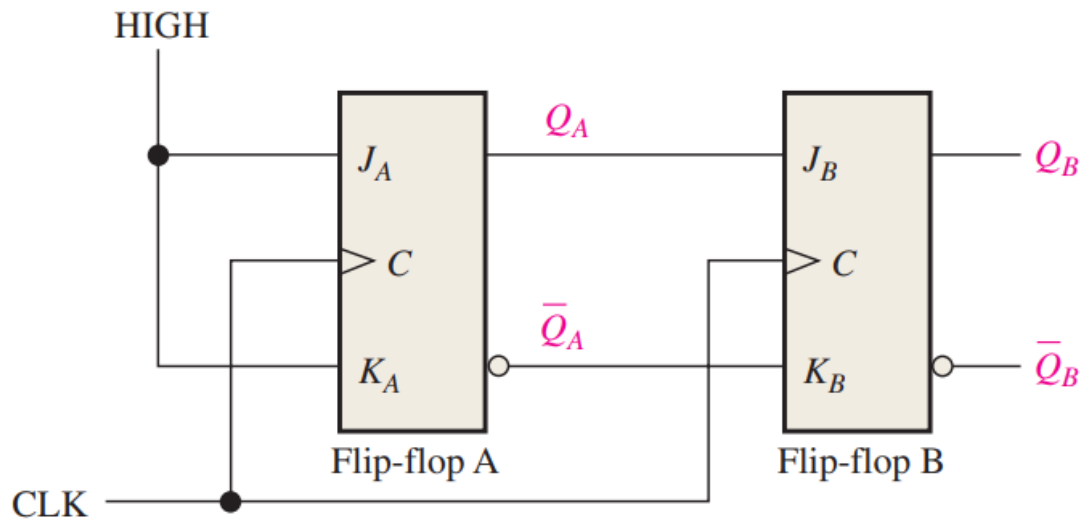
22. The flip-flop in Figure 7-88 is initially RESET. Show the relation between Q output and the clock pulse if the propagation delay t_{PLH} (clock to Q) is 8 ns.



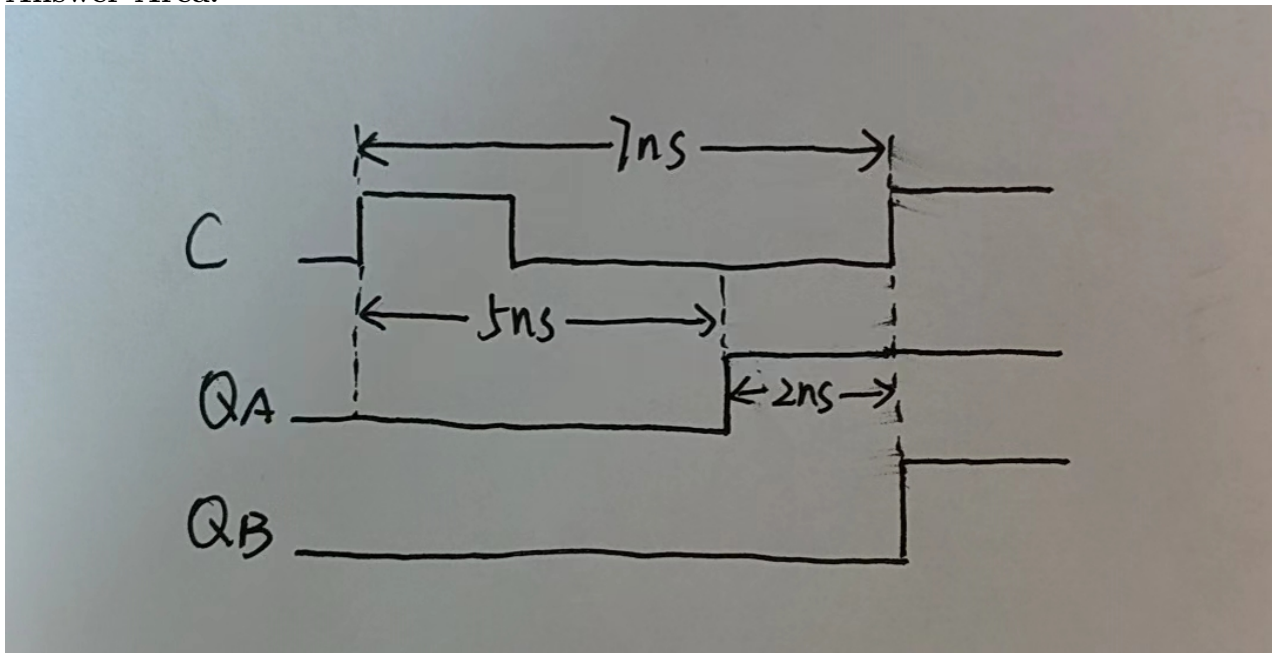
Answer Area:



24. For the circuit in Figure 7-89, determine the maximum frequency of the clock signal for reliable operation if the set-up time for each flip-flop is 2 ns and the propagation delays (t_{PLH} and t_{PHL}) from clock to output are 5 ns for each flip-flop.



Answer Area:



$$T_{min} = 2ns + 5ns$$

$$f_{max} = \frac{1}{T_{min}} = 149.2MHz$$

27. Determine the pulse width of a 74121 one-shot if the external resistor is $3.3\text{ k}\Omega$ and the external capacitor is 2000 pF .

Answer Area:

$$t_w = 0.7 \times 3.3 \times 10^3 \times 2000 \times 10^{-12} = 4.62 \times 10^{-6}s$$

28. An output pulse of $5\text{ }\mu s$ duration is to be generated by a 74LS122 one-shot. Using a capacitor of $10,000\text{ pF}$, determine the value of external resistance required.

Answer Area:

$$\therefore t_w = 0.32C_{EXT}R(1 + \frac{0.7}{R})$$

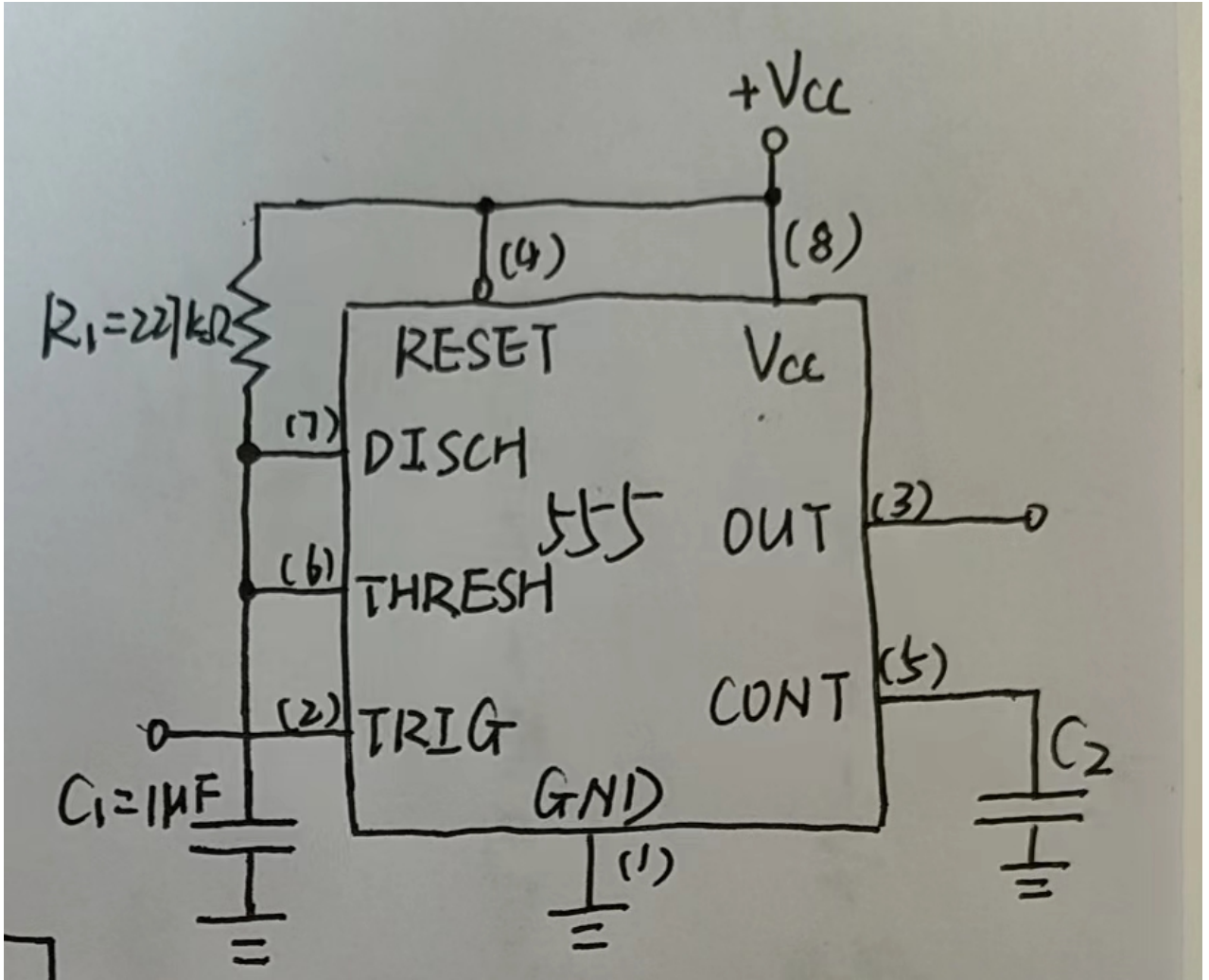
$$\therefore R = \frac{t_w}{0.32C_{EXT}} - 0.7 = \frac{5 \times 10^{-6}}{0.32 \times 10000 \times 10^{-12}} - 0.7 \times 1000 \approx 862.5\Omega$$

29. Create a one-shot using a 555 timer that will produce a 0.25 s output pulse.

$$t_w = 1.1R_1C_1 = 0.25s$$

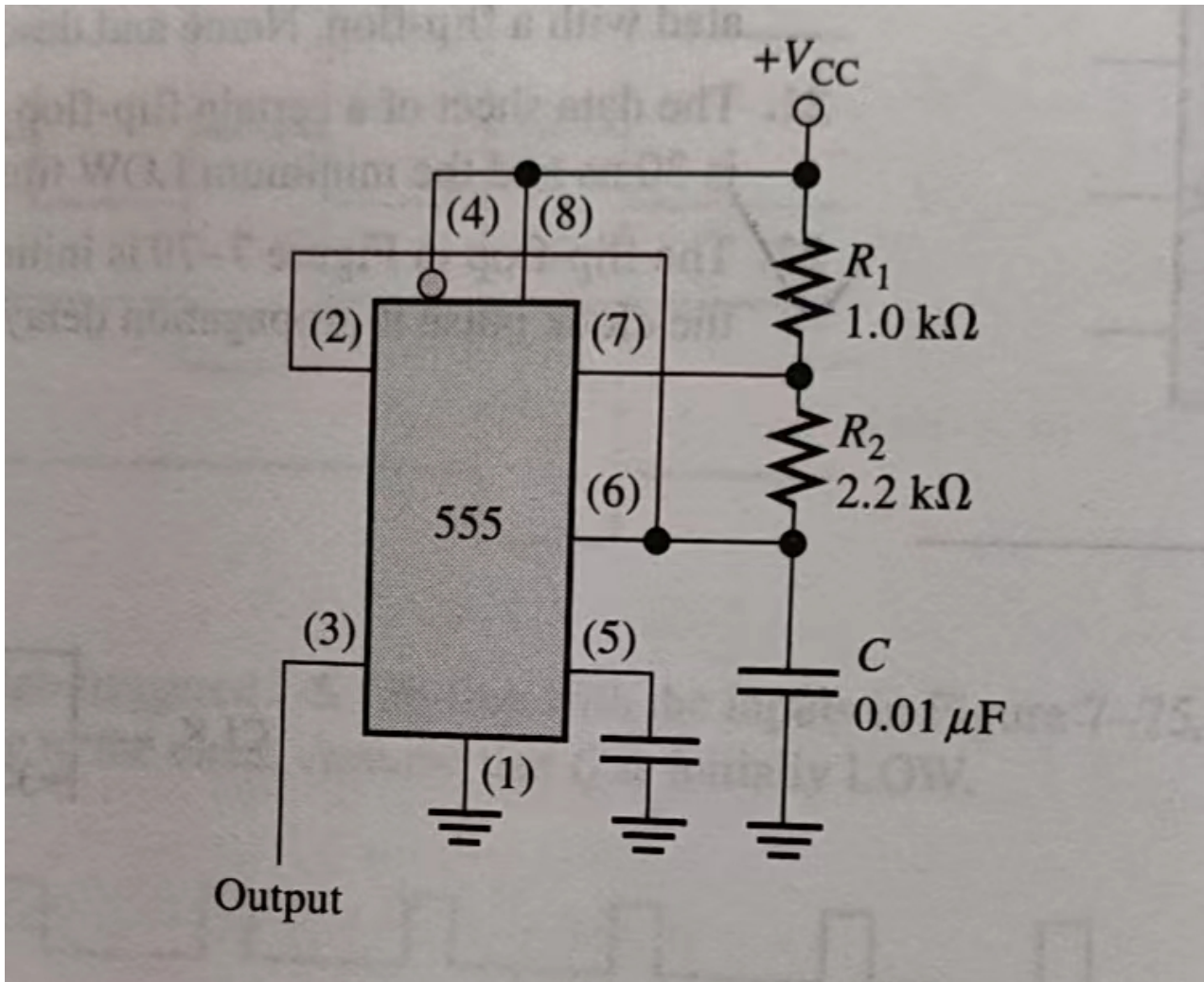
Choose $C_1 = 1\mu F$

$$R_1 = \frac{t_w}{1.1 \times C_1} \approx 227k\Omega$$



Answer Area:

30. A 555 timer is configured to run as an astable multivibrator as shown in Figure 7-91. Determine its frequency.



Answer Area:

$$f = \frac{1.44}{(R_1 + 2R_2)C} = \frac{1.44}{(1 \times 10^3 + 2 \times 2.2 \times 10^3) \times 0.01 \times 10^{-6}} \approx 26.7 \text{ kHz}$$

31. Determine the values of the external resistors for a 555 timer used as an astable multivibrator with an output frequency of 20 kHz, if the external capacitor C is 0.002 μF and the duty cycle is to be approximately 75 %.

Answer Area:

$$\begin{aligned} \therefore f &= \frac{1.44}{(R_1 + 2R_2)C} \\ \therefore R_1 + 2R_2 &= 0.36 \times 10^5 \Omega \\ \therefore \text{Duty Cycle} &= \frac{R_1 + R_2}{R_1 + 2R_2} \times 100\% \\ \therefore R_1 + R_2 &= 0.27 \times 10^5 \Omega \\ \therefore R_1 &= 18 \text{ k}\Omega, R_2 = 9 \text{ k}\Omega \end{aligned}$$

Exc1.

TO achieve a one-shot with a pulse width of approximately 10 ms, using a 74121. If select $R_{EXT} = 20 \text{ k}\Omega$, calculate the capacitance: $C_{EXT} = ?$

Answer Area:

$$\begin{aligned} \therefore t_w &= 0.7 R_{EXT} C_{EXT} \\ \therefore C_{EXT} &= \frac{t_w}{0.7 R_{EXT}} = \frac{10 \times 10^{-3}}{0.7 \times 20 \times 10^3} \approx 7.14 \times 10^{-7} \text{ F} \end{aligned}$$

Exc2:

A circuit need a 1Hz clock signal with a duty cycle of 2/3. $C = 10\mu\text{F}$. Determine the value of R_1 and R_2 .

(1) Using the first connection of 555 timer.

(2) Using the second connection of 555 timer.

Answer Area:

(1)

Handwritten calculations for the circuit:

$$f = \frac{1.44}{C_1(R_1 + 2R_2)} = 1 \text{ Hz}$$

$$\therefore R_1 + 2R_2 = 1.44 \times 10^5 \Omega$$

$$\therefore \text{Duty Cycle} = \frac{R_1 + R_2}{R_1 + 2R_2} \times 100\% = \frac{2}{3}$$

$$\therefore R_1 + R_2 = 9.6 \times 10^5 \Omega$$

$$\therefore R_1 = 48 \text{ k}\Omega \quad R_2 = 48 \text{ k}\Omega$$

(2)

